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Sensitivity Analysis of a TIR-Based Cold-Water Patch Detection Algorithm

The Influence of Cold Water Tributaries on Cold Water Patches

Bachelor Thesis

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Chapter 1

Introduction

In recent years, thermal infrared (TIR) remote sensing from uncrewed aerial vehicles (UAVs) and other airborne platforms has become an established approach for mapping thermal heterogeneity in river systems, with a growing body of work covering both data acquisition campaigns and the associated processing pipelines (Casas-Mulet et al., 2020; Dugdale, 2016; Kuhn et al., 2021; Torgersen et al., 2001). Mapping this heterogeneity is ecologically important: the cold-water patches it reveals can act as refugia that are essential for the survival of thermally sensitive fish species, particularly as river temperatures in Switzerland are expected to rise due to climate change. Despite this importance, there is currently no comprehensive information about the presence, scale, and location of such cold-water patches in Swiss water bodies. This data however is critical for river management in the context of climate change (Tonolla and Antonetti, 2021, 2026).

As part of a cantonal project aiming to quantify the temperature heterogeneity and detect cold-water patches in the Emme River and its tributaries, TIR imagery covering over a 100 km river transect were collected in a helicopter-based survey. These TIR images were then processed to extract longitudinal and transversal temperature gradients, as well as to detect cold-water spots. The current processing workflow involves stitching highly overlapping TIR images into orthophotos and performing thermal corrections using in-situ temperature loggers. Then, longitudinal and transversal temperature gradients are derived. Additionally, a newly developed algorithm for the automatic detection of cold-water patches is deployed with the goal of mapping these patches.

The current method for the automated detection of cold-water patches (CWPs) relies on externally configured parameters. Sensitivity analyses performed in (Tonolla and Antonetti, 2026) indicate that the detection results of the CWP method can vary significantly based on the chosen parameter values, with the step length and the temperature delta parameters having a particularly strong influence on the variance of the algorithm's output. The aim of this thesis is twofold: first, to systematically investigate the algorithm's variance with respect to these two parameters, with particular focus on how the presence of tributary inflows interacts with this variance. And second, to derive adjustments to the detection algorithm that could reduce this variance while maintaining ecologically plausible detection results.

The algorithm defines a CWP by comparing individual pixel temperatures against a local median reference temperature calculated over a defined segment (step length). It is hypothesised that the choice of step length introduces significant variance near the inflow of tributaries. This is because cold-water tributaries introduce sharp longitudinal temperature gradients which can skew the calculation if the step size is chosen sub-optimally.

Building on this reasoning and the twofold aim outlined above, this thesis addresses the following two research questions:

Research question 1 How does the step length contribute to the algorithm's variance in river sections containing a tributary with significantly differing water temperatures, compared to sections without such tributaries?

H_0 The variance introduced by the step length parameter is not influenced by the presence of a tributary with a substantially different temperature.

H_1 The variance introduced by the step length parameter is significantly influenced (increased or decreased) by the presence of a tributary with a substantially different temperature.

Research question 2 Building on the findings from research question 1: what adjustments could be made to the detection algorithm to reduce variability with respect to its parameters while also achieving plausible detection results?

Chapter 2

Theoretical Background

2.1 Model - Working Definition

The term model is widely used across environmental and earth sciences, yet a precise definition of a model is surprisingly challenging to find in literature. In this work, a model is understood as a set of simplified laws and processes, intertwined by dependencies, that together form a reduced reflection of an underlying system. The aim is to capture the essential behavior of that system and enable analysis from which inferences about reality can be drawn. This view is consistent with (Saltelli et al., 2007), who, drawing on Rosen (1991), describe the modelled portion of reality as an arbitrary enclosure of an otherwise open and interconnected system.

2.2 Model Input Sensitivity Analysis

The inputs to models can diver and are generally dependent in their structure and manyfoldness on the nature of the model itself. Under the light of sensitivity analysis however, it can be worthwhile to define, what the inputs to a model actually contains. In sensitivity analysis, everything that causes variation in the output of the model is classified as an input (Saltelli et al., 2007). Providing an example: If a linear regression of a target variable Y is performed, then not only are the predictors $X_1, \ldots, X_n$ considered as model inputs, but also the parameters $\theta_1, \ldots, \theta_n$ that correspond to the predictors. So the model:

$$Y = f(X_1, X_2, \theta_1, \theta_2)$$

would have four parameters form the view of sensitivity analysis.

2.3 Sensitivity Analysis

According to Saltelli et al. sensitivity analysis is "The study of how uncertainty in the output of a model [...] can be apportioned to different sources of uncertainty in the model input" (Saltelli, 2004). Sensitivity analysis provides a range of benefits. Firstly it allows the Modeler to apply a strategy called "Factor Fixing". If we discover, that the model output Y is independent of one or multiple parameters θ_i, then we can set these to fixed constants. It may then even be possible to condense a complete submodule of a model, if all parameters connected to this sub-module are noninfluential. So sensitivity analysis can be a tool for model simplification (Saltelli et al., 2007). Secondly, sensitivity analysis generally improves model validity by identifying highly influential and therefore critical model components (Ligmann-Zielinska, 2013). A range of different methods for sensitivity analysis exist. They can be grouped into global and local analysis methods.

2.3.1 Local methods

Local sensitivity analysis methods compute a sensitivity score for each parameter at a single point in the input space, typically the baseline point, which is customarily the best estimate point for the parameters. Scores are based on partial derivatives and are therefore only informative for linear models, where the derivative at one point represents behaviour across the entire input space (Saltelli and Annoni, 2010).

2.3.2 Global methods

Global sensitivity analysis methods explore the entire input space by varying all input parameters simultaneously across their plausible ranges. Sensitivity scores are therefore not tied to a single point but reflect average behaviour across the full parameter space. This makes global methods valid regardless of model linearity, as no assumption about model structure is required (Saltelli and Annoni, 2010).

2.3.3 The Morris Method

The Morris method belongs to the group of screening methods in sensitivity analysis. These are ideal to filter a short list of important input parameters from computationally expensive models, with a great number of input parameters. This can be done, as these methods only require a relatively little number of model evaluations to provide sensitivity measures for the input parameters. This makes these methods a good choice when the target of the sensitivity analysis is Factor Fixing (Saltelli et al., 2007). The following explanation of the Morris method closely follows (Saltelli et al., 2007) and is explained more explicitly in chapter 3 of their book.

Elementary Effects and the Sampling Grid

At the core of this introduction sits a k-dimensional model Y with inputs $X_1, \ldots, X_k$. These inputs all together can also be written in a vector $\mathbf{X}$. The model inputs are defined in the k-dimensional unit cube, meaning that X_i can take on a value in the interval $[0, 1]$. It is worth pointing out, that every model with a continuous input can be scaled to the $[0, 1]$ range, which makes the unit cube a great place to define methods. In the Morris method, this input space is discretized into a grid Ω. This grid features p levels.

An elementary effect of the ith input variable is then defined as:

$$EE_i = \frac{Y(X_1, X_2, \ldots, X_{i-1}, X_i + \Delta, \ldots, X_k) - Y(X_1, X_2, \ldots, X_k)}{\Delta} \qquad (3\text{-}1)$$

Where Δ represents some step in the unit cube. Δ can be chosen from a range of different values, which are given by the set $\left\{\frac{1}{p-1}, \ldots, 1-\frac{1}{p-1}\right\}$. Which value to choose for Δ from this set has implications and will be addressed in a later paragraph.

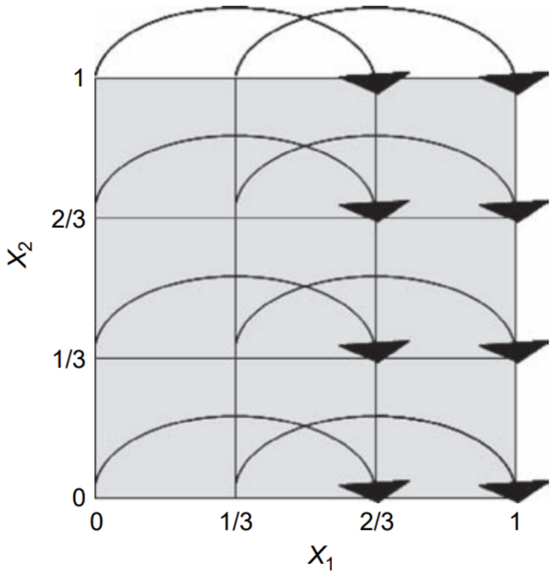

Abb. 3-1: An example grid Ω. The grid features two parameters X_1, X_2, meaning that it is a two-dimensional grid. Further, it features four levels ($p = 4$). The step length Δ chosen here is $\frac{2}{3}$, which corresponds to an element of the set of viable values for Δ: $1 - \frac{1}{4-1} = 1 - \frac{1}{3} = \frac{2}{3}$.

In the definition of the elementary effect, the vector $\mathbf{X} = (X_1, X_2, \ldots, X_k)$ has to fulfil the condition that $\mathbf{X} + \mathbf{e}_i \cdot \Delta$ still has to lie within the grid Ω. $\mathbf{e}_i$ is a vector of zeros except having a one at its ith entry. So for instance:

$$\mathbf{e}_2 = (0, 1, 0, \ldots, 0) \tag{3-2}$$

The Morris method occupies itself with finding the statistical distribution of each of the i elementary effects. $\mathcal{F}_i$ is thereby used to denote the distribution of EE_i, in other words $\mathcal{F}_i \sim EE_i$. $\mathcal{F}_i$ is a finite discrete distribution, and its number of elements depends on the number of grid levels p, which is however not that much of interest in an applied setting. A single value EE_i, however, says little on its own – how a representative sample of $\mathcal{F}_i$ is obtained is the subject of the following section (Saltelli et al., 2007).

Trajectory-Based Sampling

To populate $\mathcal{F}_i$ (and later $\mathcal{G}_i$) with multiple samples of EE_i, the Morris method uses an efficient sampling method that computes as many elementary effects EE_i with as little sampling points as possible. This is achieved by a trajectory based sampling design.

The exact mathematical formulation on how these trajectories are beeing built is beyond the scope of this thesis, a visual understanding that a trajectory contains of k + 1 points $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \ldots, \mathbf{x}^{(k+1)}$ where k again is the dimension of the unit cube. Each of the points differs form its predecessor by taking a step of Δ in one grid direction so that as a result it differs from its predecessor in only one dimension/parameter. Also the trajectory is built in a way, that at each dimension of the grid is stepped along once in the trajectory. A possible trajectory is visualized in Figure 3-2 which is taken from (Saltelli et al., 2007). Note that since the grid represented in the figure is 3 dimensional as analysis is conducted on a model with k = 3 parameters, each trajectory has 4 points.

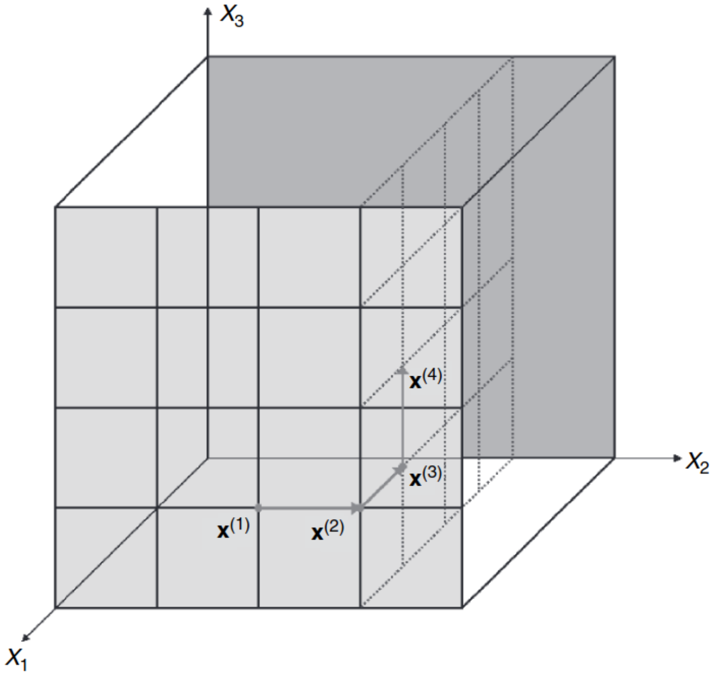

Abb. 3-2

Each trajectory yields exactly k elementary effects EE_i, one for each parameter spanning the sampling grid. A single trajectorybrequires $k + 1$ model evaluations – one at each point – and yields exactly k elementary effects: one estimate for each factor.

To build up $\mathcal{F}_i$ (and $\mathcal{G}_i$) for every factor, r such trajectories are generated. Since each trajectory contributes exactly one elementary effect per factor, each $\mathcal{F}_i$ ends up containing r samples – and it is from these r samples per factor that μ_i, σ_i, and μ_i^* are ultimately computed. The total number of model evaluations required is therefore:

$$N = r \cdot (k + 1) \tag{3-3}$$

The choice of p and Δ are not independent. When p is chosen to be even, the choice $\Delta = \frac{p/2}{p-1}$ has the additional property of guaranteeing equal-probability sampling from each $\mathcal{F}_i$, meaning that all grid levels are sampled with equal probability across the r trajectories (Saltelli et al., 2007). In the present work, $p = 6$ and jump = 3 (i.e. $\Delta = 3/5 = \frac{p/2}{p-1}$), satisfying this condition (Saltelli et al., 2007).

Sensitivity Measures: μ, σ, and μ^*

Once the distributions $\mathcal{F}_i$ have been mapped out, statistical measures can be computed just like over every other statistical distribution. In Morris, the mean μ_i and the standard deviation σ_i of $\mathcal{F}_i$ are computed. These two statistical measures together already allow for reasonable insight into the influence of the ith parameter X_i which corresponds to the ith elementary effect EE_i:

If μ_i deviates strongly from 0, then this means that the elementary effect EE_i generally takes on non-zero values and therefore that the model output changes notably when a step of Δ is performed along the dimension spanned by X_i. This then ultimately means, that changes in X_i seem to generally affect the model output Y considerably and X_i therefore receives a high sensitivity score. Generally, the more μ_i deviates from zero, the larger the sensitivity of the model output towards X_i.

If σ_i is large, this means that the distribution $\mathcal{F}_i$ is spread out considerably over a larger range. This indicates, that the elementary effects EE_i that were computed were generally different depending on their location in the unit cube. Meaning that in one area, the model might only yield small changes in output, when X_i is varied while in other areas X_i leads to strong changes in model output. This behaviour points either to nonlinearity regarding X_i or to the fact that X_i interacts with other input parameters. Which effect is responsible for the high standard deviation (or whether both are responsible for the effect at the same time) can not be determined. Which is a shortcoming of the method of Morris (Saltelli et al., 2007). Figure 3-3 provides an visual example of possible F distributions of a Factor X_i with the associated μ and σ values and the corresponding interpretation.

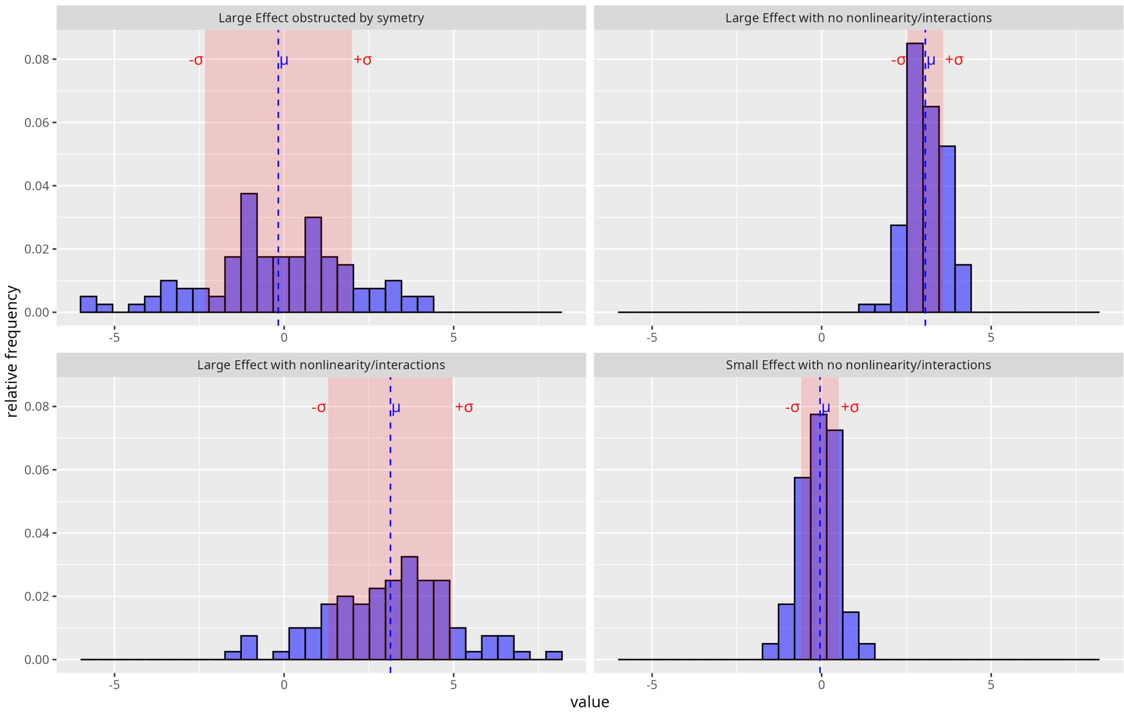

Abb. 3-3: Different F distributions of a parameter X_i are shown. Additionally, μ and σ of each distribution is provided. An interpetation is provided

Situations can occur, where $\mathcal{F}_i$ contains many extreme values however the distribution is symmetric around zero. Then computing the mean value μ_i will yield a value close to zero and we would assume, that the parameter X_i therefore is noninfluential. This would be a faulty assumption, as the sampled elementary effects EE_i are substantial in their magnitude but just cancel themselves out. This phenomenon can be spotted by always taking σ_i into consideration, as in this scenario σ_i is large, but it is more common to also sample an additional distribution $\mathcal{G}_i$ which captures the magnitude of the elementary effects $|EE_i|$. So $\mathcal{G}_i \sim |EE_i|$. We then compute the mean of this distribution μ_i^* which can be interpreted like μ_i. The further away from zero, the more influential is the parameter.

But since it is the magnitude, the distribution never features negative values and therefore cancellation effects can not occur (Saltelli et al., 2007).

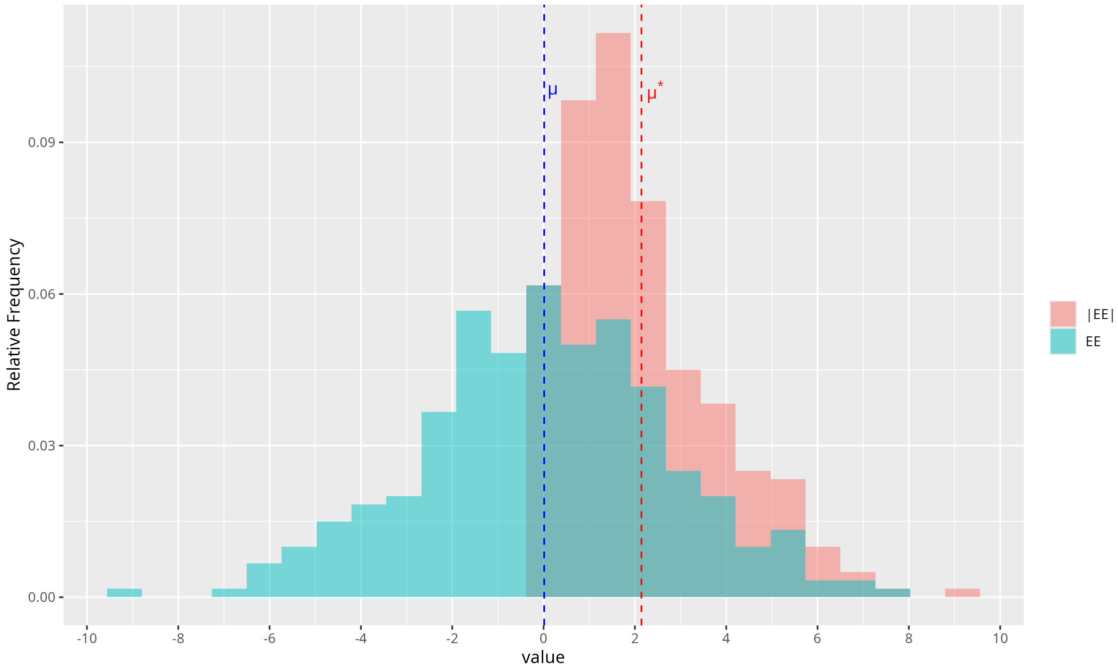

Abb. 3-4: The difference of μ and c visualized with their underlying distribution $\mathcal{G}$ and $\mathcal{F}$. While the mean μ of the distributuion $\mathcal{F}$ lies at around zero, $\mu^{\star}$ significantly deviates from zero

2.4 Coldwater Patches

Cold water patches are areas in rivers and streams that feature a colder water temperature than the temperature of the "hauptgerinne". The temperature difference varies in literature but is typically set to be between 0.5 and 3 degrees Celsius below the "hauptgerinne" temperatur, but this varies over the literature (z.B. Dugdale et al., 2013, 2015; Wawrzyniak et al., 2016; Casas-Mulet et al., 2020). Cold water patches are not automatically a cold water refugia. These are coldwater patches which are also used by coldwater organisms for thermal relief (Meja et all.). The importance of cold water refugia is expected to rise with the effects of climate change (IPCC, 2014)

Chapter 3

Methods

This chapter describes the operationalisation of the research questions into a set of methodological steps, which are then described in detail. The presented methods can be divided into three logical steps. The first step focuses on optimizing the currently available CWP detection algorithm with respect to execution speed, to set the stage for the sensitivity analysis. The second step focuses on deriving sensitivity indices using the Morris method for a set of variable CWPs which can be clearly attributed to being either tributary-caused or non-tributary-caused. The third step focuses on deriving visualizations and fitting a statistical model to the selected CWPs, aiming to explain the sensitivity of CWP area to the step length parameter as a function of CWP area and the temperature difference to the surrounding environment.

3.1 Data

3.2 Operationalisation of the Research Question

Defining river sections containing a CWP is not entirely straightforward. The general location of such a river section can be determined easily using a map, as the point at which the tributary enters the river must lie within it. However, no general rule exists for determining the precise spatial extent of such a section – that is, where it begins and ends. For example, it could be defined as extending 50 meters upstream from the tributary inflow, or as a section downstream of the inflow whose length scales with the tributary's discharge, under the assumption that larger tributaries influence the thermal regime of the receiving river over a longer distance.

In this study, an outcome-based definition of a tributary-influenced river section was used instead: rather than geometrically delineating river sections and applying a subjective rule for their extent, the river section is defined directly as the extent of the CWP caused by the tributary. This removes the ambiguity of delineating tributary-influenced river sections, since such a section is then simply spanned by the CWP itself. This definition also inherently captures the requirement of a "significantly differing water temperature" from research question 1: if a tributary's water were not significantly colder than the receiving river, no CWP would form. The existence of a tributary-caused CWP therefore provides evidence that the tributary's temperature differs significantly from that of the surrounding river.

This operationalisation, however, narrows the scope of research question 1 to tributaries that are colder than the receiving river: a tributary with significantly warmer water would not produce a cold-water patch, and such cases are therefore not captured by this approach, even though research question 1 does not explicitly exclude them. This consequence of the chosen operationalisation is discussed further in the Limitations section.

Analogously, river sections without a tributary inflow are defined simply as the extent of the corresponding non-tributary-caused CWP, ensuring a consistent spatial definition is applied across both classes.

3.3 Optimizationn of Existing Algorithm

Most sensitivity analysis methods feature a high cost when it comes to model executions. Morris is one of the cheaper methods but still requires a reasonable number of model executions: a model with $k = 3$ parameters and $r = 50$ requires $N = 50 \times (3 + 1) = 200$ executions (Equation 3-3). To perform these executions within a feasible timespan for the provided CWP algorithm, optimisation of the computations is needed. Additionally, parallel execution of the algorithm should be performed to further reduce execution times for the sampling. For the latter, ZHAW's High Performance Cluster (HPC) is used, which has direct implications on the optimisation approach. To ensure that the code-optimisation speed gains also carry over to the infrastructure (e.g., the underlying distributed file system) of the HPC, the optimisation and benchmarking are performed directly on the HPC system. In the remainder of this thesis, the original implementation of the algorithm is referred to as v1, and the resulting speed-optimized reimplementation as v2.

3.3.1 Reimplementation of the CWP algorithm based on GDAL and system calls

The original algorithm focuses on per-slab processing of the raster using a set of R loops and relying on `terra` and `sf` functions to implement the logic. The reworked algorithm adapts much of the original logic but instead operates directly on disk: The majority of processing steps read their input from, and writes their output to, raster or vector files via calls to the GDAL command line interface. Table 3-1 gives a high-level overview of the reimplemented algorithm; the tool or function used in each step is listed in the right-hand column. The remainder of this section motivates the key design decisions underlying this reimplementation.

Tab. 3-1: High-level overview of the reworked CWP detection algorithm, including the primary tool or function used for each step.

Step	Description	Tool / Function
1	Divide the river reach into n centerline segments; construct a wide zone polygon and a narrow reference-strip polygon for each segment.	`sf::st_buffer`
2	Rasterize the zone polygons to a raster matching the TIR raster resolution, using the segment id as burn-in value.	`gdal vector rasterize`
3	Round the TIR raster using a fixed rounding factor.	`gdal raster calc`
4	Extract the median temperature per segment from the rounded TIR raster within each reference strip; store in a look-up table keyed by segment id.	`exactextractr::exact_extract`
5	Reclassify the zone raster, replacing each segment id with its reference temperature from the look-up table, yielding the reference raster.	`gdal raster reclassify`
6	Subtract the rounded TIR raster from the reference raster; flag pixels where the difference exceeds ΔT, producing a binary CWP raster.	`gdal raster calc`
7	Polygonize the binary CWP raster into patch polygons.	`terra::as.polygons`
8	Filter patches by minimum area, compute per-patch temperature statistics, and apply a patch-level threshold check against the reference temperature.	`exactextractr::exact_extract`

Performance. This decision was made as `terra` operations are sometimes not performant, and GDAL offers an often faster set of geocomputational tools, which can be called from R using `system()` or `sf::gdal_utils()` (Lovelace et al., 2019). Empirical testing in a prior coursework exercise also showed a significant speedup of GDAL-based raster operations compared to `terra` across a range of use cases (Dürig, 2026); this coursework was not peer-reviewed and is cited here only as a supporting empirical observation.

Memory footprint and HPC suitability. The GDAL-based approach also enables easy chunk processing with intermediate reading and writing to disk, which helps to keep memory footprint small. This can be beneficial in an HPC setting, where not only the number of available cores but also the available memory can become a limiting factor. The drawback of this decision is that writing intermediate results to disk causes large spikes in disk space consumption, as intermediate rasters of the algorithm can be up to 25 GB in size. This was counteracted by deploying compression during writing and decompression during reading – a strategy that produces little computational overhead but greatly reduces the size of the intermediate results. Specifically, this was implemented by setting the compression options `COMPRESS=DEFLATE` and `PREDICTOR=2` when working with GDAL's GeoTIFF driver (GDAL/OGR contributors, 2026). It should be noted, however, that heavily relying on GDAL introduces a large amount of I/O activity, which can significantly hurt performance on distributed file systems; this remains a limitation of the chosen approach and is revisited in the Discussion Section.

Raster value extraction. As a further optimization, `terra::extract()` was replaced with `exact_extract()` from the `exactextractr` package (used in steps 4 and 8 of Table 3-1). This was done as `exact_extract()` extracts aggregated values from rasters considerably faster than `terra::extract()` for large rasters (Mieno, 2022); it should be noted, however, that this speed difference was not tested in isolation within this work. A further difference between the two functions concerns how partially covered cells are handled: `exact_extract()` performs a coverage-fraction-weighted extraction, in which partially covered cells are weighted by their fraction of coverage (ISciences, LLC, 2026), whereas `terra::extract()` by default includes a cell only if its center point falls within the polygon (Hijmans et al., 2024) which is the behaviour used in the original (v1) implementation. To approximate this cell-center-inclusion behaviour within `exact_extract()`, only cells with a coverage fraction of at least 50% were included; this approximation is not exactly equivalent to `terra`'s center-point rule, and its consequences for output equivalence between v1 and v2 are addressed in the Discussion Section.

3.3.2 Output eqivalence and Runtime Comparison

The resulting reimplementation was tested for equivalence in output by comparing the produced CWP polygons to the CWP polygons of the old algorithm when deployed with the exact same set of parameters. As a tool for comparison visual analysis in QGIS and the computation of Jaccard similarities between the two sets of CWPs was deployed. Table 3-2 shows the parameter combinations deployed in testing. Further, the runtime between the new algorithm and the existing version was compared with respect to the step length parameter. Runtime was compared across three segment lengths (400, 800, and 1200 m), with all other parameters held constant (zone halfwidth = 60 m, buffer = 3 px, ΔC = 1.0 °C, minimum patch area = 2 m^2, rounding = 0.1 °C, 8-connectivity enabled).

Tab. 3-2: Parameter combinations used for the output equivalence comparison between v1 and v2.

Parameter	Test 1	Test 2	Test 3
`segment_length_m`	200	800	1200
`zone_halfwidth_m`	60	60	60
`buffer_px`	4	3	5
`delta_C`	0.5	1.0	1.5
`min_patch_area_m2`	2	3	2
`round_to`	0.1	0.1	0.1
`connect_diagonals`	TRUE	TRUE	TRUE

3.4 Sensitivity Analysis using Morris

With the algorithm optimized, sensitivity analysis can now be carried out to determine the sensitivity of the CWP detection algorithm with respect to the parameters buffer pixel count, step length, and temperature delta. Especially of interest is to observe the difference in sensitivity towards the step length parameter between tributary-caused and non-tributary-caused CWP. For this reason, a pre-selection method is deployed to guarantee that the studied CWP are sensitive towards the step length parameter.

The inclusion of buffer pixel count and temperature delta alongside step length also serves a secondary, validatory purpose. (Tonolla and Antonetti, 2026) already found in their sensitivity analysis that the buffer pixel count seems to have little influence on the CWP area; if this finding can be replicated here, algorithmic simplification using factor fixing might be possible. Conversely, the temperature delta parameter was among the most influential parameters in (Tonolla and Antonetti, 2026). Replicating both findings within the present methodology would lend additional credibility to the approach used here, as well as the approach deployed by (Tonolla and Antonetti, 2026).

Figure 4-1 provides a high-level overview of the methodology described in this section. The general approach can be logically segmented into three steps. The first step focuses on determining CWP that are susceptible to the step length parameter and classifying these CWP into tributary-caused and non-tributary-caused CWP. The second step focuses on deriving Morris sensitivity indices for these CWP with respect to their area. The third step focuses on visualizing these sensitivity indices and fitting a statistical model relating the sensitivity of CWP area to the step length parameter to the CWP's area and its temperature difference to the surrounding environment.

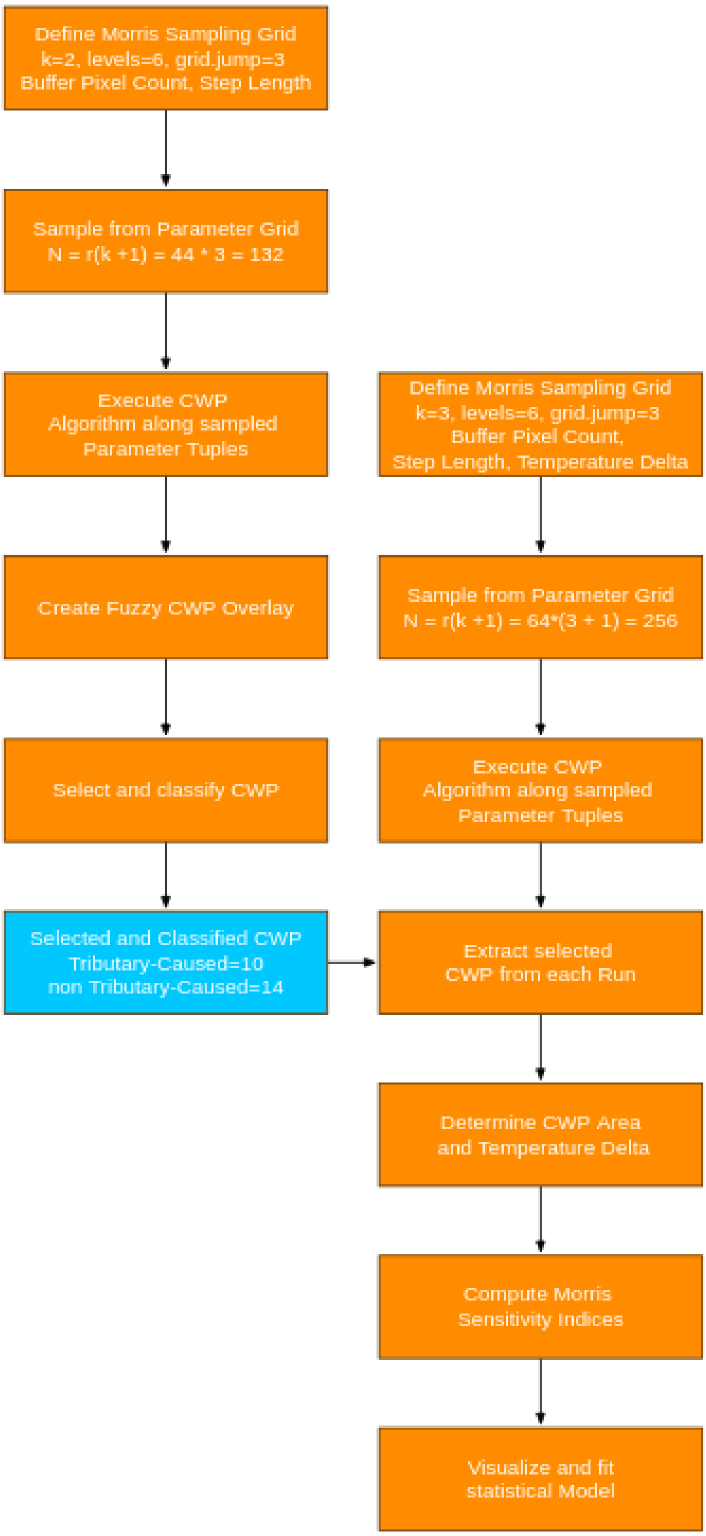

Abb. 4-1: High-level overview of the Morris sensitivity analysis methodology, comprising three stages: (1) determining and classifying step-length-sensitive CWP (Section **??**), (2) computing Morris sensitivity indices with respect to CWP area (Section **??**), and (3) visualizing these indices and fitting a statistical model relating sensitivity to CWP area and temperature delta (Section **??**).

3.4.1 Determining and classifying step length sensitive CWP

First, a Morris design with $r = 44$ and $k = 2$ is set up for the buffer pixel count and step length parameters. Then the speed-optimized (v2) CWP algorithm is deployed on the resulting parameter tuples. Lastly, the results are aggregated to a fuzzy representation of the CWP over all runs, to then select CWP which are variable with respect to the CWP algorithm and to classify them into tributary-caused and non-tributary-caused CWP.

Setting up first Morris experiment

First, a two-dimensional Morris sampling grid was set up spanning two parameters: buffer pixel count and step length. A total of 6 levels were set for each parameter, with the step length varying from 400 m to 1200 m and the number of pixels in the buffer spanning from 2 to 7. To compute the elementary effects, a jump distance of 3 was set. Figure 4-2 shows the deployed grid visualized in greater detail, with the jump distance additionally drawn in.

From this grid, a total of 44 trajectories were sampled ($r = 44$), resulting in a total of $44 \times (2 + 1) = 132$ parameter tuples being generated. Each tuple holds a different combination of buffer pixel count and step length. These were written to a file along with a unique identifier for each tuple. For the sampling procedure, the R package `sensitivity` was used.

The decision to vary only the buffer pixel count and step length in this first Morris experiment (i.e., $k = 2$, with the temperature delta parameter held constant) is methodologically important. The temperature delta parameter directly determines the threshold used by the algorithm to flag pixels as part of a CWP; varying it would therefore be expected to introduce variability in the detected extent of essentially all CWPs, independent of whether they are sensitive to the step length parameter specifically. Including the temperature delta in this first step would consequently make it difficult to distinguish CWPs that are variable because of the step length parameter from CWPs that merely appear variable because the CWP-defining threshold itself was varied. By holding the temperature delta constant and varying only step length and buffer pixel count, any variability observed in the resulting fuzzy overlay can be more confidently attributed to these two parameters, and in particular to step length (as buffer pixel generally is non influential).

A Morris design was used for this first step, rather than e.g. a simple grid sweep, to ensure the validity of the selection criterion itself: the goal of this step is to select CWPs whose variability can subsequently be quantified using the Morris sensitivity indices computed in the second experiment. Had a different sampling scheme been used here, it would be possible to visually identify CWPs as variable based on a pattern of variation that the Morris elementary effects might not register as sensitive. By already constructing the fuzzy overlay from Morris-sampled runs in this first step, any variability visible in the overlay is, by construction, of the kind that the subsequent Morris-based sensitivity analysis is designed to detect.

Model Evaluation along Parameter Combinations

Next, a parallelized computation scheme was set up to evaluate the CWP algorithm along the created parameter tuples, while keeping the parameters not part of the Morris ex-

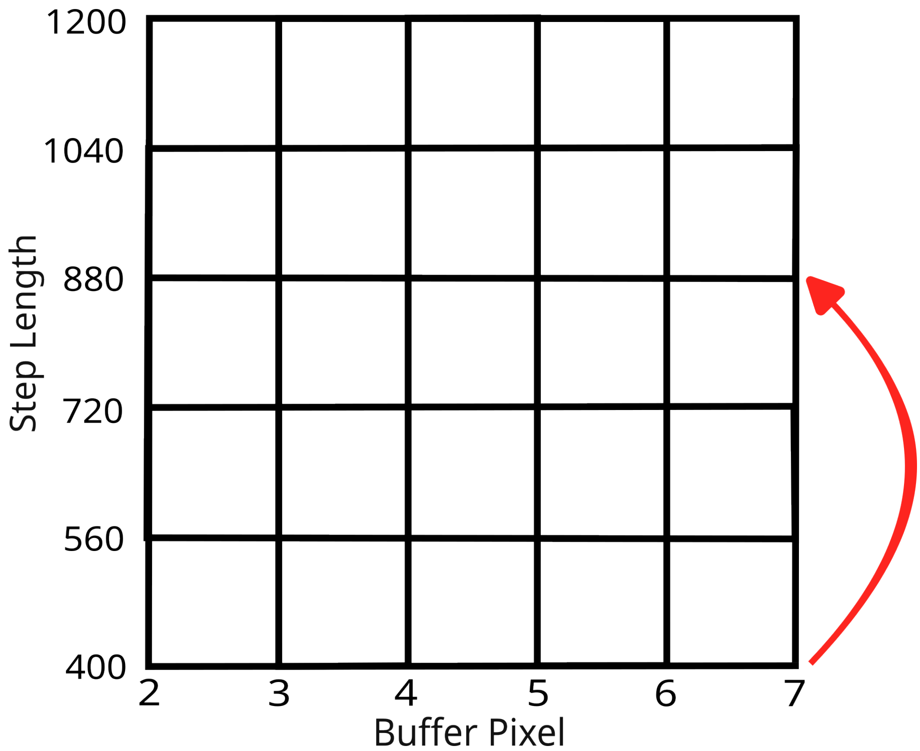

Abb. 4-2: Visual depiction of the Morris sampling grid, spanning step length and buffer pixel count.

periment constant. Table 4-3 shows the values at which the other parameters were held constant.

Parameter	Value
`zone_halfwidth_m`	60 m
`delta_C`	1.0 °C
`min_patch_area_m2`	2 m^2
`connect_diagonals`	TRUE
`round_to`	0.1

Tab. 4-3: Parameter values held constant during the first Morris experiment (Section 3.4.1), while `step_length` and `buffer_px` were varied according to the sampled parameter tuples.

To orchestrate the parallel execution of the CWP algorithm, the open source software GNU Parallel was deployed (Tange, 2024). Specifically, GNU Parallel was used to implement a worker pool of processes, each executing the CWP algorithm with a parameter tuple fetched from a parameter text file. Before starting the computation, each process also sets up an individual working directory, to which intermediate results as well as the final result are written. To keep the parameter combination as well as the identifier traceable – and the results therefore identifiable – the folder name holds both the identifier and the parameter values.

Figure 4-3 visualizes the GNU Parallel workflow.

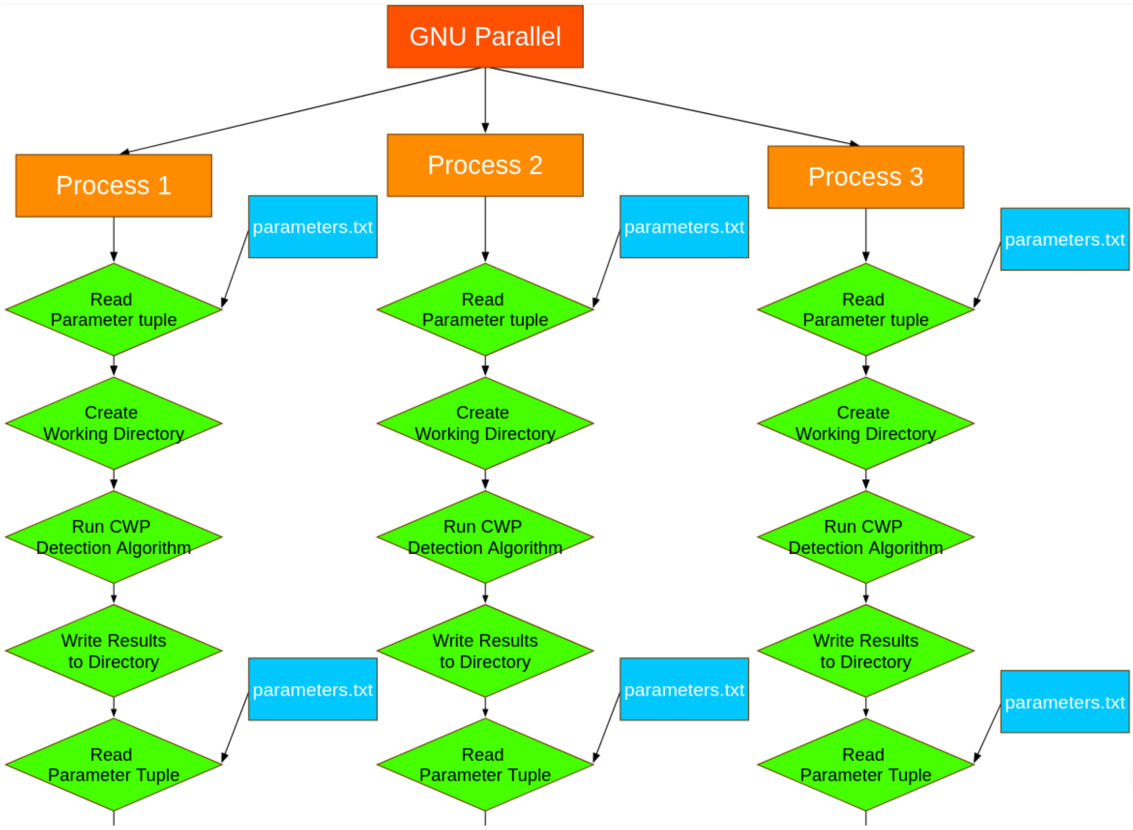

Abb. 4-3: TODO: describe what this plot shows.

Creation of a fuzzy overlay of CWP

The resulting CWP polygon layers were then processed into a single raster through a series of processing steps. First, each polygon layer was rasterized using the same pixel size as the TIR rasters. Then, these different raster representations of the CWPs were stacked into a multi-layer raster, and a pixel-wise mean calculation was performed. This aggregates the three-dimensional raster stack into a two-dimensional raster. The pixel-wise mean computation produces a pixel score between 0 and 1, which can be interpreted as follows: a pixel value of 1 indicates that this area of the Emme basin was part of a CWP during each of the runs, while a pixel value near zero indicates that the area covered by the pixel was almost never part of a CWP in the model outputs. The resulting raster can be understood as a fuzzy membership representation of the CWPs in the Emme, where each pixel has a probability-like score of belonging to a CWP. Figure 4-4 shows this fuzzy overlay raster.

From these fuzzy CWPs, it can also be read which CWPs are variable with respect to the step length (or the buffer pixel count – which, however, has a negligible influence, as seen in the results). CWPs which show a wide range of different shades in their coloring contain pixels that are generally variable in their identity as CWP or non-CWP pixels. These CWP are therefore candidates for the selection and classification process described in the following subsection.

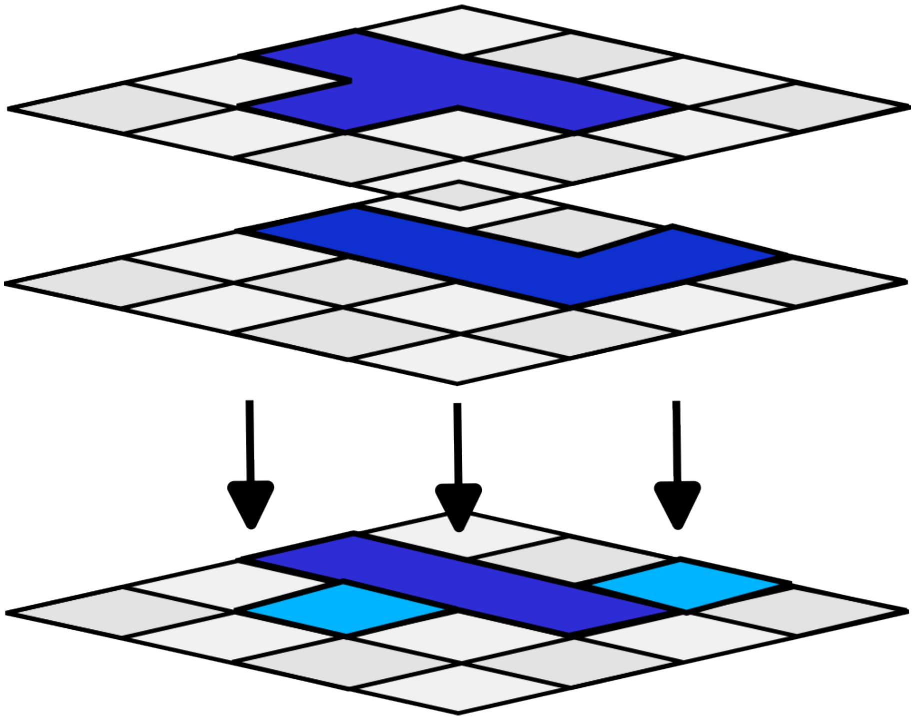

Abb. 4-4: Fuzzy overlay raster representing the pixel-wise mean across all CWP detection runs. Pixel values range from 0 (never part of a detected CWP) to 1 (part of a detected CWP in every run), and can be interpreted as a fuzzy membership score for each pixel.

Selection and Classification of CWP

From this fuzzy CWP raster, CWP which were clearly variable and whose origin could be clearly attributed to either a tributary inflow or the absence thereof were selected and classified into the groups tributary-caused and non-tributary-caused. The selection and classification process was not carried out algorithmically due to time constraints, but was based on a visual interpretation of the patches. First, patches were judged as being variable or non-variable based on their coloring in the fuzzy overlay. Patches were generally determined to be variable when their coloring contained a substantial proportion of grey and black pixels, indicating that the detected CWP extent changed considerably across runs. These CWP then underwent classification. A CWP was classified as tributary-caused if it was located in close proximity to a tributary inflow – either directly at, or immediately downstream of, the point where the tributary enters the receiving river.

To identify tributary inflows, OpenStreetMap as well as SwissImage were consulted. OpenStreetMap facilitated the identification of tributaries which might be obscured by forest or buildings in the SwissImage data, while SwissImage provided the opportunity to spot tributaries which might not be mapped in OpenStreetMap, as OpenStreetMap is based on volunteered geographic content and therefore cannot be assumed to be complete.

A total of 24 CWP patches which are variable and can be clearly attributed to being either tributary-caused or non-tributary-caused were selected and classified. Per the operationalisation introduced in Section **??**, each of these 24 CWP and its spatial extent constitutes one of the river sections referred to in research question 1. For each selected CWP, a bounding box was extracted. Figure 4-5 illustrates the classification process.

Multible small adjacent CWPs where grouped together and counted as beeing one CWP patch. This was done by having spatial autocorrelation in mind. Especially, Tobler's first law of geography states that "all things are related but close things are more related than distanced things." Small CWP close together near the same location share the smale local generating conditions (same tributary, same reach geometry, same local thermal regime). Their sensitivity to the step length parameter will therefore be highly similar precisely because of their spatial proximity. Sampling them all as seperate CWP would consequently over-represent that location in the analysis, as if that spot were sampled multiple times and counted as inependent variation. This would skew the analysis towards these local conditions.

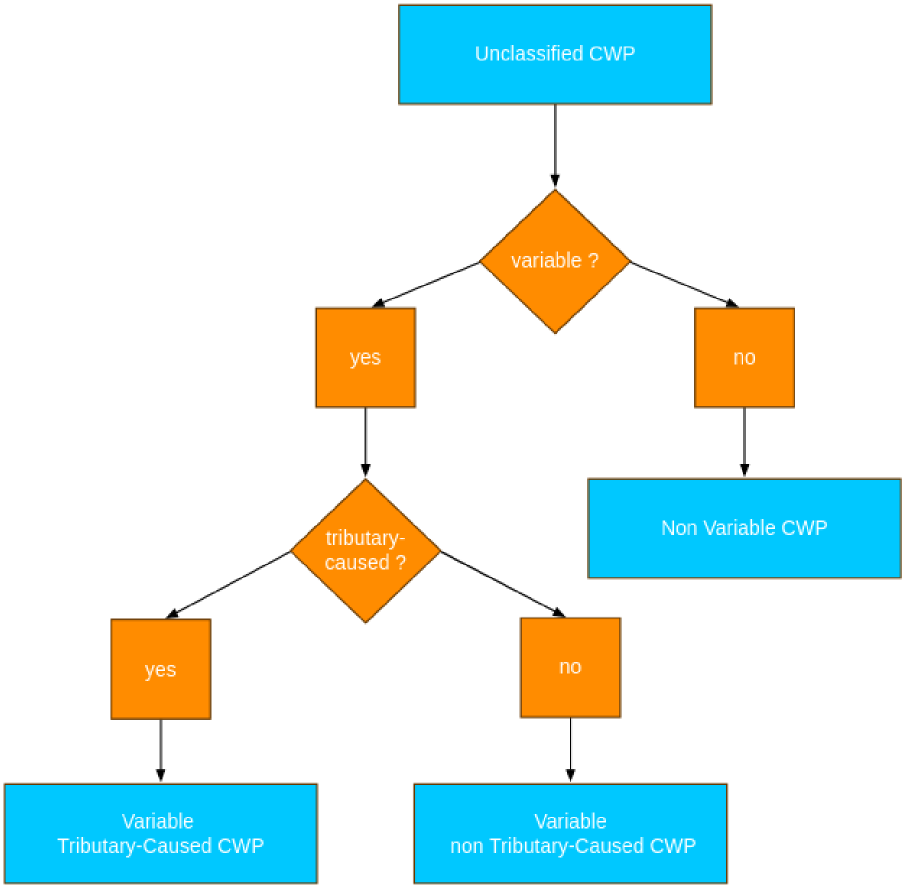

Abb. 4-5: Decision tree illustrating the visual classification process applied to candidate CWP patches: patches are first assessed as variable or non-variable based on their coloring in the fuzzy overlay, and variable patches are further classified as tributary-caused or non-tributary-caused based on their proximity to a tributary inflow.

3.4.2 Computing morris sensitivity indices with respect to CWP area

Setting up second sampling Grid and executing Algorithm

with these cwps classified, a second morris experiment was set up this time also varying the temperature delta parameter next to the buffer px and the step length.A visual depiction of the second morris sampling grid is provided in Figure xzy. A total of 64 trajectories where sampled form this grid creating a total of $64 * (3 + 1) = 256$ sampling tupleds. Then the model was again executed in prallel on the different parameter tuples with the remaining parameter held constant, which are provided in table xyz For the execution orchestration, the same set up as in 3.4.1 was deployed. The resulting cwp polygons where again written to folders uniquely identified by an id, aswell as the parameter values.

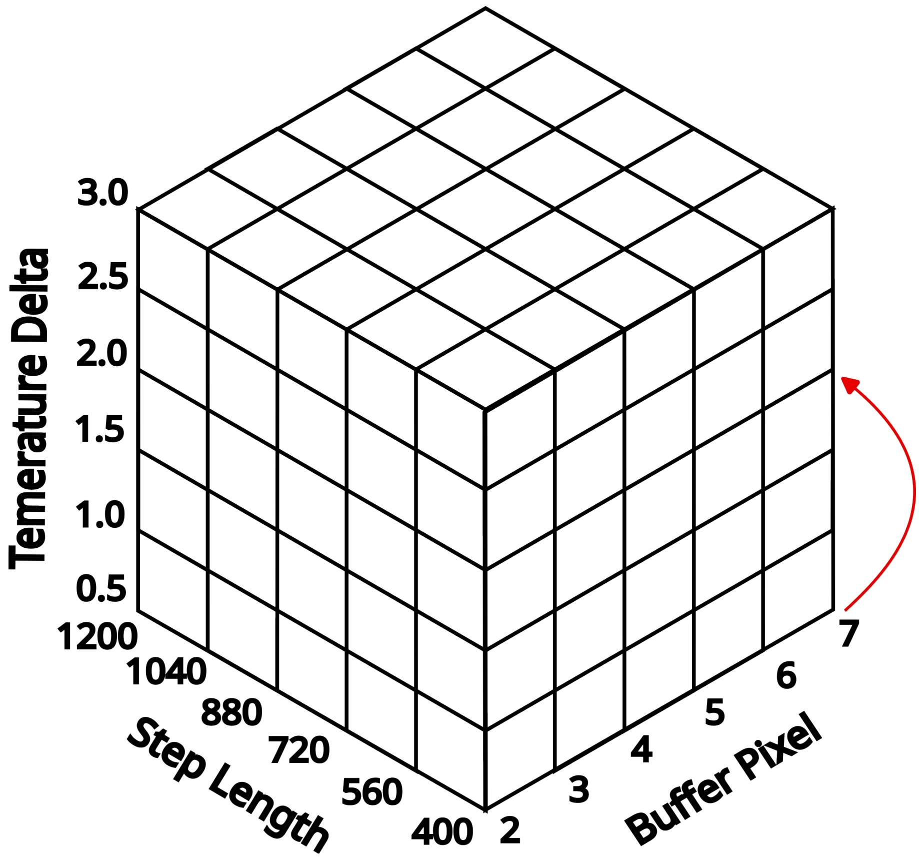

Abb. 4-6: Visual depiction of the second Morris sampling grid, spanning step length, buffer pixel count, and the temperature delta threshold.

Computing lumped statistics for the selected CWPs

In the next step, the bounding boxes of the 24 selected CWPs and the polygon outputs from the second Morris experiment are brought together. For each bounding box and each Morris run, the CWP polygons lying inside the bounding box are extracted. The total area covered by CWP within the bounding box and the area-weighted mean temperature delta across all CWP polygons inside it are then computed. Figure 4-7 summarizes this per-location, per-run computation schematically.

Two special cases arise in this computation. First, if a bounding box contains only a single CWP polygon, the area-weighted mean temperature delta simply corresponds to the temperature delta of that polygon. Second, if a bounding box is completely empty for a given run – which occurs when the algorithm produces no CWP at that location for the corresponding parameter combination – both the total area and the temperature delta are set to zero. Setting the area to zero is straightforward: no detected CWP is equivalent to a CWP with zero surface area. Setting the temperature delta to zero is equally justified: a CWP is defined by a temperature difference to the surrounding river exceeding a given threshold; a location with no detected CWP is therefore treated as having no thermal anomaly, making a temperature delta of zero the natural neutral value.

The result of this process is a table in which each row corresponds to one annotated CWP

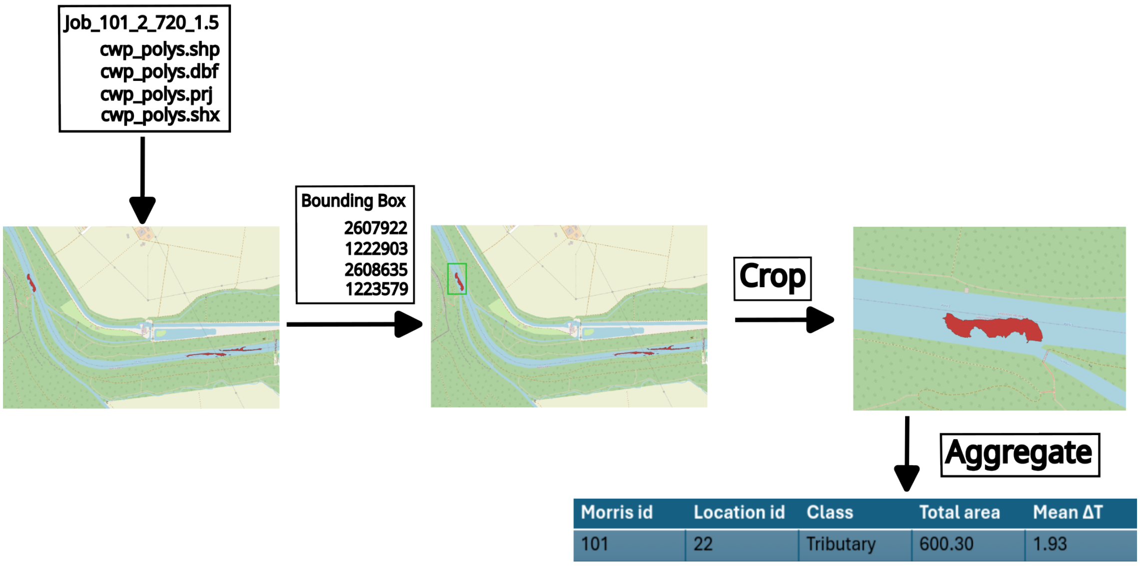

Abb. 4-7: Schematic overview of the lumped-statistics computation for one annotated CWP location (Limpbach inflow into the Emme near Utzenstorf BE). For each Morris run, CWP polygons are cropped to the bounding box; if polygons are present, total area and area-weighted mean temperature delta are computed; if the bounding box is empty, both are set to zero.

location under one Morris parameter tuple, with the total detected CWP area and mean temperature delta as the two output measures. Table 4-4 shows an excerpt of this table.

Location	Class	ΔT [°C]	Total CWP area [m^2]
2	Non-tributary	3.67	32.32
2	Non-tributary	4.13	22.65
2	Non-tributary	4.28	18.92
⋮	⋮	⋮	⋮
4	Tributary	2.40	39.82
4	Tributary	2.91	15.89
4	Tributary	0	0
⋮	⋮	⋮	⋮

Tab. 4-4: Excerpt of the lumped statistics table. Each row corresponds to one annotated CWP location under one Morris parameter tuple. The area measures differ across rows sharing the same location, reflecting the variability of the detected CWP extent with respect to the input parameters.

As the table illustrates, the area measures generally differ for the same CWP location across different parameter tuples. The total CWP area is therefore a suitable scalar statistic for characterising the variability of each CWP with respect to the input parameters.

Computing Morris sensitivity indices

The Morris sensitivity indices μ, μ^*, and σ are then computed with respect to the total CWP area for each bounding box. Before computation, the total CWP area values are normalised per location using min-max scaling into the $[0, 1]$ range. This is necessary because without normalisation, locations with large CWP would produce larger absolute area differences than locations with small CWP, simply due to their size – making the resulting sensitivity indices incomparable across locations. After scaling, an area value of 1 corresponds to the largest area ever produced for that bounding box across all runs, and a value of 0 corresponds to an empty bounding box.

The sensitivity indices are then computed with respect to these normalised area values using the `sensitivity` package, by supplying it with the saved Morris design object from the second experiment and the computed area measures. As introduced in the theoretical background, Morris produces one sensitivity index per parameter that was varied in the experiment. With $k = 3$ parameters (step length, buffer pixel count, and temperature delta) and two indices of interest (μ^* and σ), this yields $3 \times 2 = 6$ sensitivity values per location. Across all 24 annotated locations, this produces a total of $24 \times 6 = 144$ sensitivity values, which form the basis of the analysis presented in the following section.

3.5 Visualisation and Statistical Modeling

Two sets of boxplot visualisations were produced. The first displays the distributions of μ^* and σ with respect to the step length parameter, stratified by CWP class (tributary-caused and non-tributary-caused), to allow a direct visual comparison of sensitivity between the two groups. The second displays the μ^* distribution with respect to all three varied parameters – buffer pixel count, step length, and temperature delta – pooled across both classes, to provide an overview of the relative influence of each parameter on CWP area.

Additionally, a linear model was fitted to gain insight into whether the mean area of a CWP and its mean temperature delta to the surrounding environment are related to its sensitivity to the step length parameter. The response variable was μ^* with respect to step length, and the two predictors were the mean total CWP area across all Morris runs for that bounding box and the mean temperature delta across all Morris runs for that bounding box. The model was fitted using R's `lm()` function, i.e.:

$$\mu^*_{\text{step length}} = \beta_0 + \beta_1 \cdot \overline{\text{area}} + \beta_2 \cdot \overline{\Delta T} + \varepsilon \tag{5-1}$$

The model was fitted across both classes pooled rather than separately per class.

3.6 Usage of KI

The LLM Claude Sonnet 4.6 and Claude Opus 4.6 (Antrophic) was used as an aid to ensure proper formulation, spelling and grammatical correctness. Further it was used to derive the LaTeX for Images and simulation protocol table. The models where also used as a coding assitant

Chapter 4

Results

4.1 Comparison of outputs

The three comparison runs achieved outputs that are in general agreement with respect to CWP location and shape, but are not identical at the pixel level. The Jaccard similarities for the three runs are 0.936, 0.956, and 0.976, indicating generally good agreement between v1 and v2.

Figure 1-1 shows the CWPs generated during trial run 3, with polygons of the speed-optimized algorithm (v2) shown in red and polygons of the original algorithm (v1) shown in blue. The two algorithms are in general agreement, however do not fully agree at the pixel level. Figure 1-2 shows a more pronounced example of this pixel-level disagreement for trial run 1, while Figure 1-3 shows an example of perfect agreement between the two methods.

Abb. 1-1: CWPs generated during trial run 3 (Jaccard similarity = 0.976). Polygons of the speed-optimized algorithm (v2) are shown in red, and polygons of the original algorithm (v1) are shown in blue. The two algorithms are in general agreement, but do not fully agree at the pixel level.

Abb. 1-2: Example of more pronounced pixel-level disagreement between v1 and v2 for trial run 1 (Jaccard similarity = 0.936). Polygons of v2 are shown in red, polygons of v1 in blue.

Abb. 1-3: Example of complete spatial agreement between v1 and v2 for trial run 2 (Jaccard similarity = 0.956). Where polygons overlap perfectly, only the red v2 polygons are visible.

4.2 Speed comparison in execution

Figure 2-4 shows the execution time of v1 and v2 as a function of segment length. The execution time of v2 is approximately 0.5 hours across all three tested segment lengths, remaining largely constant. In contrast, v1 shows execution times ranging from approximately 2 hours at a segment length of 400 m to approximately 1.25 hours at 1200 m, with execution time increasing as segment length decreases (i.e., as the number of segments increases).

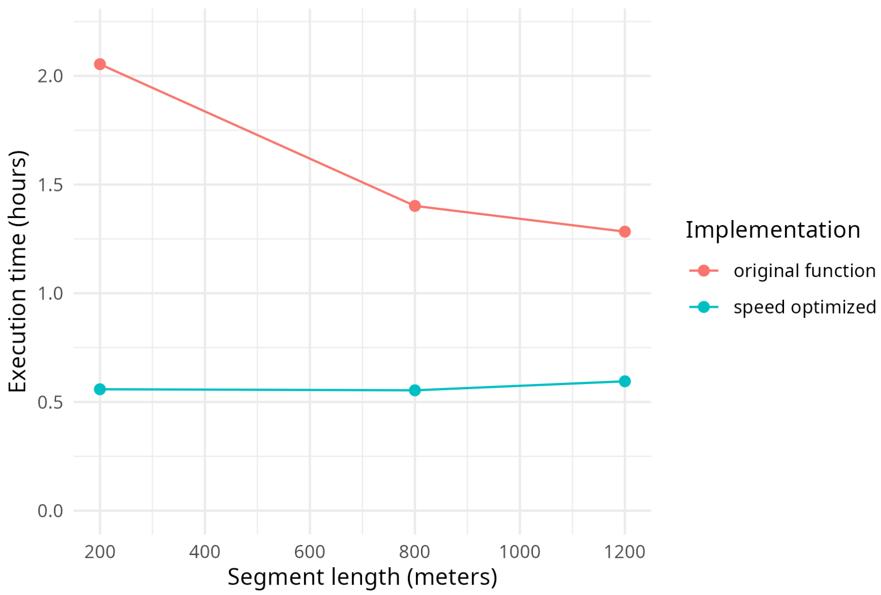

Abb. 2-4: Execution time of the original (v1) and optimized (v2) implementations of `detect_cwp_single` as a function of segment length. The optimized implementation shows execution times that are both shorter and largely constant across segment lengths, whereas the original implementation scales disproportionately with decreasing segment length (i.e., increasing number of segments).

4.3 Sensitivity scores for step length parameter

Figure 3-5 shows the Morris sensitivity indices μ^* and σ with respect to the step length parameter, stratified by CWP class ($n = 10$ tributary-caused, $n = 14$ non-tributary-caused). All non-tributary-caused CWP show μ^* values below 0.2, indicating low sensitivity to the step length parameter across this class. The tributary-caused CWP show a wider spread in μ^*: some tributary-caused CWP have sensitivity values comparable to the non-tributary-caused group, while others show substantially higher values of up to 0.3–0.6. The median μ^* values of the two classes are similar, with the primary difference lying in the upper tail of the tributary-caused distribution. The σ values show a comparable pattern, with greater spread in the tributary-caused group.

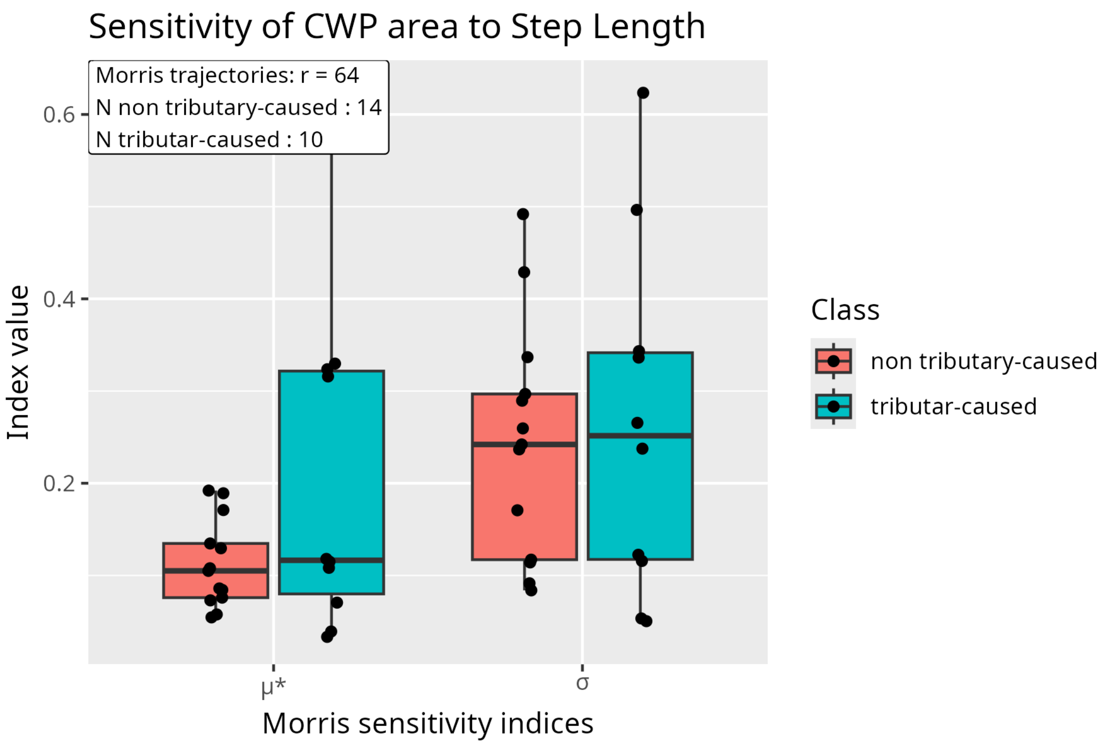

Abb. 3-5: Morris sensitivity indices (μ^* and σ) for the `step_length` parameter with respect to cold water patch area, stratified by patch class (red: tributary-caused, $n = 10$; blue: non-tributary-caused, $n = 14$). Boxplots show the distribution across bounding boxes.

4.4 Overall μ^* across all parameters

Figure 4-6 displays the μ^* values for all 24 CWP locations with respect to the three parameters varied in the second Morris experiment. The buffer pixel count shows consistently low μ^* values across all locations, with the entire distribution concentrated near zero. In contrast, both step length and temperature delta show distributions that deviate clearly from zero. The temperature delta parameter shows the highest μ^* values overall, suggesting it has the strongest influence on the detected CWP area across the set of annotated locations.

4.5 μ^* (step length) versus CWP area

Figure 5-7 shows the 24 selected CWP along three dimensions simultaneously: mean CWP area averaged across all Morris runs (x-axis), μ^* with respect to the step length parameter (y-axis), mean temperature delta to the surrounding environment averaged across all Morris runs (color), and CWP class (shape). The data suggest a positive relationship between mean CWP area and sensitivity to the step length parameter: locations with larger mean CWP area tend to show higher μ^* values. This pattern appears across both classes. No clear relationship between mean temperature delta and μ^* is visible from the coloring.

Table 5-1 shows the results of the linear model fitted to predict μ^* (step length) from mean CWP area and mean temperature delta. The mean area predictor is highly significant ($p < 0.001$) with a positive slope, consistent with the visual pattern in Figure 5-7. The mean temperature delta predictor is not significant ($p = 0.556$), indicating no detectable linear

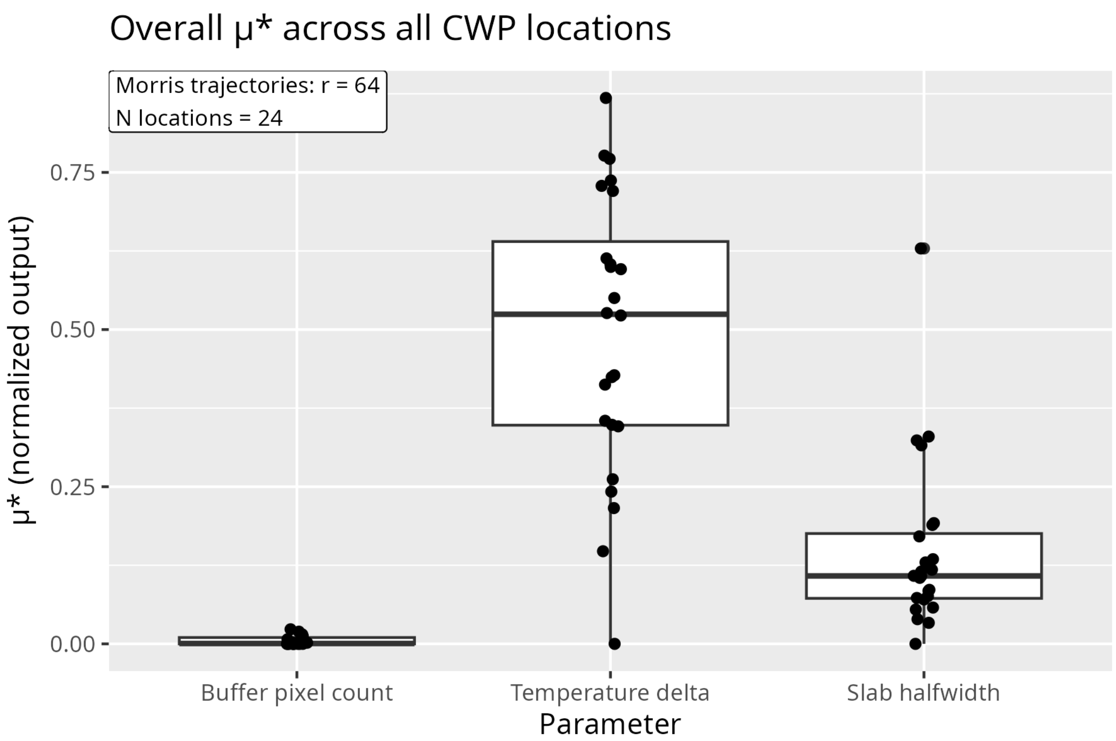

Abb. 4-6: Overall μ^* sensitivity indices for `buffer_px`, `delta_T`, and `step_length` across all classified cold water patches ($N = 24$ locations, $r = 64$ trajectories).

relationship between temperature delta and sensitivity to the step length parameter after accounting for area. The model explains a substantial proportion of variance ($R^2 = 0.611$, Adjusted $R^2 = 0.572$).

Tab. 5-1: Regression results for μ^* (step length) predicted by mean CWP area and mean ΔT, fitted across all 24 annotated locations.

Predictor	Estimate	SE	t	p
(Intercept)	9.024×10^{-2}	3.094×10^{-2}	2.92	0.009
Mean area	1.578×10^{-4}	2.839×10^{-5}	5.56	< 0.001
Mean ΔT	-1.041×10^{-2}	1.739×10^{-2}	-0.60	0.556

Note. $R^2 = 0.611$, Adjusted $R^2 = 0.572$, $F(2, 20) = 15.73$, $p < 0.001$. Residual standard error = 0.088 on 20 degrees of freedom (one observation deleted due to missingness).

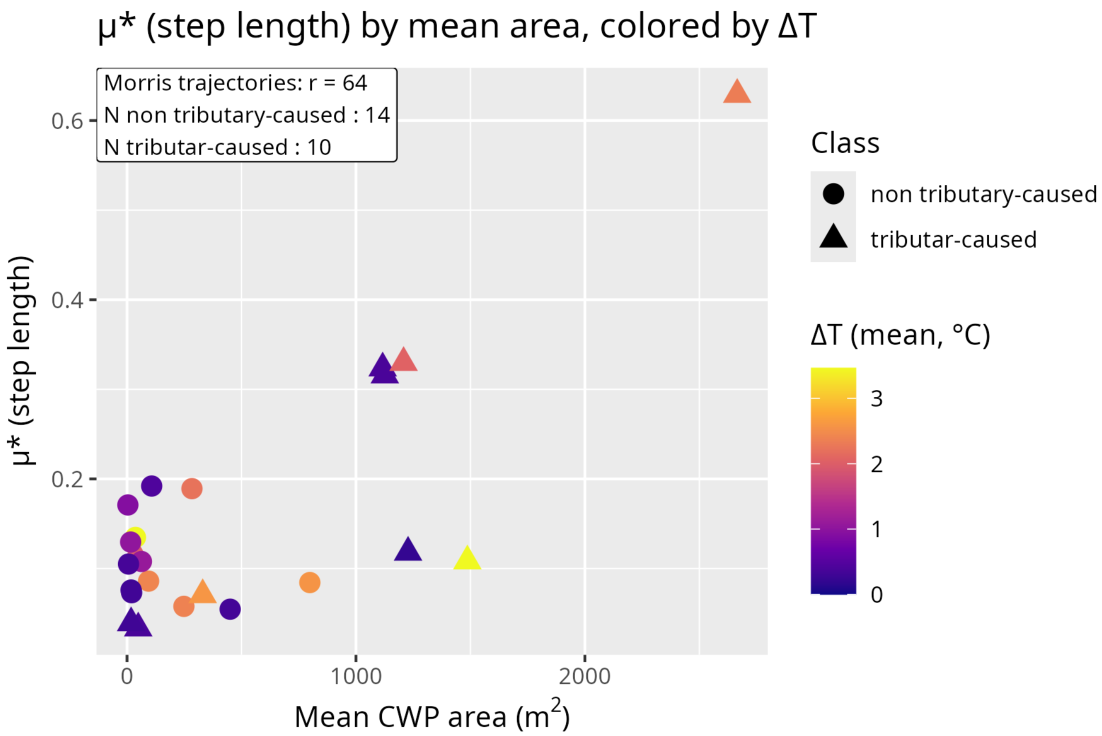

Abb. 5-7: μ^* (step length) versus mean CWP area for each of the 24 annotated locations. Color indicates the mean temperature delta to the surrounding river ($\overline{\Delta T}$, averaged across all Morris runs); shape indicates CWP class (circle: non-tributary-caused; triangle: tributary-caused).

Chapter 5

Algorithm Modification based on Morris Screening Results

5.1 Insights from the Morris Screening

The Morris sensitivity analysis yielded three actionable findings with respect to the parameters of the CWP detection algorithm. First, the buffer pixel count was found to be non-influential with respect to detected CWP area, with μ^* values close to zero across all 24 annotated locations and low σ values indicating an absence of meaningful interactions with the other parameters. This justifies the application of factor fixing: the buffer pixel count is internalized and set to a constant value of 3, which lies within the range of the Morris screening grid, effectively removing it as a free parameter. Second, the step length parameter was found to have a non-negligible influence on detected CWP area, with the mechanistic analysis suggesting that this sensitivity arises from the possibility of a CWP spanning a slab boundary and therefore being compared against inconsistent reference temperatures. This motivates a redesign of the reference temperature computation that does not rely on a fixed slab length. Third, the temperature delta parameter was found to be the most influential of the three parameters. This parameter cannot be removed, as different temperature delta thresholds are used in the literature to define CWPs, and the user must therefore retain the ability to set it according to their chosen definition.

Taken together, these findings motivate a modified algorithm, referred to hereafter as v3, that removes the buffer pixel parameter by factor fixing and replaces the slab-based reference temperature computation with an approach that does not depend on a step length parameter.

5.2 Operationalisation of Surrounding Stream Temperature

The step-length dependency was addressed by rethinking the operationalisation of the surrounding stream temperature used as the reference for pixel flagging. In v1 and v2, the reference temperature for each pixel is determined by the median temperature of the slab segment to which that pixel belongs. This approach has two consequences: it requires a user-defined step length, and it produces a reference temperature field that is piecewise constant – uniform within each slab and discontinuous at slab boundaries. CWPs that happen to span a slab boundary are therefore compared against two different reference temperatures, which can produce artificial cutoffs in the detected patch extent.

In v3, this is replaced by an inverse distance weighting (IDW) scheme. Rather than dividing the reach into fixed slabs, reference temperature sample points are obtained by sampling multiple step length values from a uniform distribution and extracting the median temperature at each resulting segment's visual center. These sample points, pooled across all sampled step lengths, are then used to compute a spatially continuous reference temperature raster via IDW, where each cell's reference temperature is a distance-weighted average of the surrounding sample point temperatures. This guarantees that the reference temperature is a spatially continuous field, and that no sharp cutoffs at slab boundaries can occur.

5.3 Implementation Detail

The implementation of v3 follows the structure of v2 closely, differing only in the computation of the reference temperature raster (steps 1–5 below, replacing steps 1–5 of Table 3-1). The full reference temperature computation proceeds as follows:

1. From a uniform distribution with bounds 500–3000 metres, sample 20 candidate step length values.
2. For each of the 20 step length values:
   (a) Divide the river reach into n centerline segments based on the step length.
   (b) Construct a narrow 3-pixel-wide reference strip polygon for each segment.
   (c) Extract the median temperature of each segment from the (unrounded) TIR raster using the reference strip polygon.
   (d) Compute the visual center of each strip polygon and assign the corresponding median temperature to this point.
3. Pool the reference points (location + temperature) from all 20 step length values into a single combined set of IDW sample points.
4. Initialise a new raster to be populated with reference temperature values.
5. For each cell in this raster:
   (a) If the corresponding cell in the TIR raster is `NaN`, set the cell to `NaN` and proceed to the next cell.
   (b) Otherwise, compute a weighted temperature value for this cell using IDW over the pooled reference points.
6. Round the TIR raster using a fixed rounding factor.
7. Subtract the rounded TIR raster from the reference raster; flag pixels where the difference exceeds ΔT, producing a binary CWP raster.
8. Polygonize the binary CWP raster into patch polygons.
9. Filter patches by minimum area, compute per-patch temperature statistics, and apply a patch-level threshold check against the reference temperature.

Step 5b is implemented in C++ and imported into R via the `Rcpp` package, as the per-pixel IDW computation was identified as the primary computational bottleneck.

5.4 Reduction of Parameters

The modifications described above result in a reduction of the algorithm's free parameters. The buffer pixel count is fixed internally at a value of 3. The step length parameter is no longer required, as the IDW reference temperature computation samples reference points across a range of step lengths rather than relying on a single user-defined value. The zone halfwidth parameter is likewise no longer needed, as the extent of the reference raster is matched to the extent of valid pixels in the TIR raster. It should be noted that this implies a constraint on the input data: v3 can only be applied to TIR rasters that contain an exact water mask of the river channel, with non-river pixels set to `NaN`.

The remaining free parameters of v3 are: the paths to the TIR raster and the river centerline, the minimum patch area, the rounding factor, the temperature delta threshold, and whether diagonal pixels are counted as neighbours during polygonization.

5.5 Evaluation of the Modified Algorithm

5.5.1 Output comparison

The CWP polygons produced by v3 were compared against those of v1 using the same Jaccard similarity metric applied in the output equivalence comparison of v1 and v2 (Section 3.4.1). The Jaccard similarities obtained were 0.325, 0.334, and 0.287 for the three validation parameter combinations – substantially lower than the values of 0.936, 0.956, and 0.976 obtained for v2 versus v1. This indicates that v3 produces CWP detections that differ considerably from those of v1, which is expected: the IDW-based reference temperature computation represents a fundamental redesign of the core reference temperature step rather than a minor reimplementation, and is therefore not expected to replicate v1's outputs closely. The relevant question is not whether v3 agrees with v1, but whether v3's detections are ecologically more plausible.

Figure **??** illustrates the most direct qualitative improvement. In v1, the detected CWP extent is truncated at the slab boundary: the patch ends abruptly where one slab meets the next, not because the underlying thermal anomaly ends there, but because the reference temperature changes discontinuously at that point. In v3, the spatially continuous IDW reference temperature surface removes this artificial boundary, allowing the detected patch to extend naturally beyond the location of the former slab boundary and taper off gradually as the temperature difference to the surrounding river diminishes.

A direct consequence of eliminating slab-boundary cutoffs is that CWPs which were previously reported as two separate patches – one on each side of a slab boundary – are now detected as a single, larger polygon. Figure **??** shows an example of this: two fragments present in v1 are merged into one contiguous CWP in v3, which is likely a more accurate representation of the underlying thermal anomaly.

Finally, v3 was found to detect CWPs at several locations where neither v1 nor v2 produced any detection. Figure **??** shows one such example. A likely explanation is that the corresponding thermal anomaly straddles a slab boundary in v1 and v2, causing it to be detected as two sub-threshold fragments that are subsequently removed by the minimum area filter. In v3, the absence of slab boundaries allows the anomaly to be detected as a single contiguous patch above the area threshold. Whether these additional detections represent genuine CWPs missed by v1 and v2, or false positives introduced by the IDW reference temperature surface, cannot be determined without independent ground-truth validation – but the former interpretation is consistent with the mechanistic argument developed in this chapter.

5.5.2 Runtime

The v3 algorithm has an execution time of approximately 35 minutes, comparable to that of v2. As with v2, this runtime is expected to remain largely constant across different parameter combinations, since neither the minimum patch area, the rounding factor, nor the temperature delta threshold have a substantial effect on the volume of computation performed by the algorithm.

Bilder/idw_cutoff_eliminated.png

Abb. 5-1: Elimination of an artificial slab-boundary truncation. Left: v1 output, showing the CWP extent truncated abruptly at a slab boundary due to the discontinuous piecewise-constant reference temperature field. Right: v3 output, showing the patch extending naturally beyond the former boundary location, with the detected extent tapering gradually in accordance with the underlying temperature gradient.

Bilder/idw_merged_cwp.png

Abb. 5-2: Merger of previously severed CWP fragments. Left: v1 output, showing two separate CWP polygons on either side of a slab boundary. Right: v3 output, showing the same thermal anomaly detected as a single contiguous patch.

Bilder/idw_new_cwp.png

Abb. 5-3: Example of a CWP detected by v3 but absent in v1 and v2. The detection is consistent with the hypothesis that the corresponding thermal anomaly was split into sub-threshold fragments by a slab boundary in v1 and v2, and subsequently removed by the minimum area filter.

Chapter 6

Discussion

6.1 Difference in CWP polygons between v1 and v2

The CWP polygons produced by v1 and v2 are in general agreement with respect to location and spatial extent, as reflected by Jaccard similarities of 0.936, 0.956, and 0.976 across the three validation runs. The observed discrepancies are confined to the pixel level – typically affecting a small number of pixels at patch boundaries – and no systematic spatial bias is apparent.

The most likely source of these discrepancies is the difference in how `terra::extract()` and `exactextractr::exact_extract()` handle partially covered pixels: as described in the Methods, the 50% coverage-fraction threshold used to approximate `terra`'s center-point inclusion rule is not exactly equivalent, and small differences in the extracted reference temperatures could propagate to pixel-level differences in the binary CWP raster. Importantly, the differences between v1 and v2 are minor and bounded in their effect: they do not introduce systematic differences in which CWPs are detected or where, but only affect a small number of boundary pixels.

The research question is specifically targeted at the behaviour of v1. The sensitivity analysis was, however, conducted on v2 as this optimized algorithm version has a substantially reduced algorithm runtime. The high Jaccard similarities, combined with the observation that the discrepancies between v1 and v2 can be attributed to a minor rounding artefact rather than to any fundamental difference in algorithmic behaviour, support the conclusion that v2 replicates the behaviour of v1 to a degree sufficient to transfer sensitivity findings. In other words, the sensitivity of CWP area to the step length, buffer pixel count, and temperature delta parameters as observed in v2 can be considered representative of the sensitivity that would have been observed had the Morris experiment been conducted on v1 directly.

6.2 Speed comparison

The optimized algorithm (v2) shows substantially shorter execution times than v1 across all three tested segment lengths, with the performance advantage being most pronounced at shorter step lengths. Of particular interest is the contrasting scaling behaviour of the two implementations: while v2 maintains an approximately constant execution time of around 0.5 hours regardless of step length, v1's execution time increases as step length decreases, ranging from approximately 1.25 hours at 1200 m to approximately 2 hours at 400 m.

This difference in scaling behaviour is consistent with a mechanistic explanation rooted in how the two implementations process the river reach. v2 is loop-free: GDAL operates on the full raster in a single pass, and each pixel undergoes a fixed number of operations regardless of how many segments the river reach is divided into. The number of segments therefore has no bearing on execution time, which explains the observed constant scaling behaviour. v1, by contrast, processes the reach one slab at a time using R loops. Each loop iteration incurs both a fixed overhead – the cost of the loop machinery itself, independent of slab size – and a variable computation proportional to the raster area of that slab. At shorter step lengths, the reach is divided into a larger number of smaller slabs: the variable computation per iteration decreases while the fixed overhead remains constant, shifting the ratio of overhead-to-useful-work unfavourably and increasing total execution time. This is a plausible explanation for the observed scaling behaviour of v1, though it should be noted that this mechanistic account is a hypothesis and was not formally tested within this work.

The decision to optimize the algorithm prior to the Morris experiment was well-founded: v2 is faster than v1 across the entire parameter range tested, and the performance advantage is largest precisely at the shorter step lengths (400 m) where v1 is slowest – which happen to fall within the Morris sampling grid used in this work.

6.3 Answering research question 1

The boxplot comparison of μ^* values between tributary-caused and non-tributary-caused CWP shows a tendency consistent with H1: tributary-caused CWP display a wider spread in step-length sensitivity, with a higher third quartile and higher maximum values than the non-tributary-caused group. However, a closer inspection reveals that this pattern is not uniform within the tributary-caused class — some tributary-caused CWP show sensitivity values comparable to the non-tributary-caused group, while others are substantially more sensitive. This within-class heterogeneity suggests that tributary presence alone is not a sufficient condition for high sensitivity to the step length parameter, and that H1 as stated does not fully capture the situation. At the same time, the data do not support H0 either: the non-tributary-caused CWP consistently show low μ^* values, and the elevated sensitivity observed in a subset of tributary-caused CWP is unlikely to be entirely coincidental. The true picture lies between the two hypotheses.

The scatter plot and linear model together point to CWP area as the more proximate driver of step-length sensitivity. Across both classes, locations with larger mean CWP area tend to show higher μ^* values, and this relationship is approximately linear. The linear model confirms this: the mean area predictor is highly significant ($p < 0.001$) with a positive slope, while the mean temperature delta predictor is not significant ($p = 0.556$), indicating no detectable independent contribution of temperature delta to step-length sensitivity after accounting for area. The color distribution in the scatter plot is consistent with this: both high and low ΔT CWP appear across the full range of μ^* values, with no clear separation by temperature.

A mechanistic explanation for the area–sensitivity relationship follows naturally from the structure of the algorithm. CWPs in a river system tend to be elongated rather than wide, since the river channel constrains lateral extent while longitudinal extent is less bounded. Large CWP are therefore predominantly long CWP. A long CWP has a higher probability of spanning the boundary between two adjacent slabs, and when this occurs, the two parts of the patch are compared against different reference temperatures – one computed from each slab's reference strip. If those reference temperatures differ (which is more likely near a tributary inflow, where sharp longitudinal temperature gradients exist), the flagging threshold is effectively applied inconsistently across the patch, introducing variability in the detected CWP extent that is directly attributable to the step length parameter. Shorter or smaller CWP are more likely to fall entirely within a single slab, and are therefore less exposed to this effect.

Tributary inflows are then best understood not as a direct cause of step-length sensitivity, but as one mechanism that can produce large, elongated CWP – by introducing substantial volumes of cold water that generate extended thermal anomalies along the river. This explains why tributary-caused CWP tend to show higher sensitivity as a group, while also explaining why the effect is not universal: a tributary-caused CWP that is small (perhaps because the tributary discharge is low or the temperature difference modest) will not be sensitive to the step length parameter for the same reason that a small non-tributary-caused CWP is not. It also means that the data cannot speak to whether a large non-tributary-caused CWP would show comparable sensitivity to an equally large tributary-caused CWP, since no non-tributary-caused CWP larger than approximately 1000 m^2 is present in the dataset.

Interestingly, CWP area and mean temperature delta appear to be largely unrelated in the data, despite the intuitive expectation that a colder tributary might produce a larger thermal anomaly. This could suggest that tributary discharge plays a more important role than temperature difference in determining CWP size – a large, moderately cold tributary may produce a larger CWP than a small but very cold one. Discharge data were not collected during the survey, but could be a valuable addition in future work.

The following answer to research question 1 is therefore proposed: neither H0 nor H1 is fully supported by the data. H0 is too strong a rejection – tributary presence is associated with a tendency towards higher step-length sensitivity. However, H1 overstates the role of tributary presence as a direct determinant of sensitivity. The data suggest that the more proximate driver is CWP size: large CWP, which are disproportionately but not exclusively tributary-caused, are more susceptible to the step length parameter because they are more likely to span slab boundaries and therefore be exposed to inconsistent reference temperatures.

Two important caveats must accompany this conclusion. First, the sample size of $n = 24$ is small, and the significant area–sensitivity relationship could reflect a chance finding. Second, and more fundamentally, the selection and classification of the 24 CWP was carried out through visual interpretation rather than algorithmically, which introduces the possibility of confirmation bias: CWP were selected on the basis of appearing variable and appearing tributary-caused or non-tributary-caused, meaning the dataset reflects the analyst's prior expectations rather than a random or representative sample of CWP in the Emme system. For these reasons, the findings should be treated as hypothesis-generating rather than confirmatory, and replication with a larger, algorithmically selected sample would be needed before stronger conclusions can be drawn.

6.4 Overall Sensitivity to the Different Parameters

The overall μ^* distributions across all 24 annotated locations reveal a clear ranking of parameter influence with respect to detected CWP area. The buffer pixel count is consistently non-influential, with μ^* values close to zero across all locations. The corresponding σ values are similarly low, suggesting that the elementary effects of buffer pixel count are both small in magnitude and consistent across the parameter space – indicating neither a strong main effect nor meaningful interactions with the other parameters. This finding is consistent with (Tonolla and Antonetti, 2026), who report the same result in their sensitivity analysis of the CWP detection algorithm. Given this replication, the strategy of factor fixing can be applied: the buffer pixel count can be set to a fixed constant value, effectively removing it as a free parameter of the algorithm. The specific value to which it is fixed is discussed in Chapter 6.

The temperature delta parameter shows the highest μ^* values overall, indicating that it is the most influential of the three parameters with respect to CWP area. This is also consistent with the findings of (Tonolla and Antonetti, 2026), who likewise identify temperature delta as the dominant parameter. The elevated σ values for temperature delta suggest that its elementary effects vary across the parameter space, pointing to the presence of nonlinearity or interactions with the other parameters. This is ecologically plausible: the relationship between the flagging threshold and the detected CWP area is unlikely to be linear, as small changes in temperature delta near the boundary of what constitutes a detectable thermal anomaly could produce disproportionately large changes in detected area, while changes far from this boundary may have little effect.

The step length parameter occupies an intermediate position, with μ^* values clearly above zero but consistently below those of temperature delta. Like temperature delta, step length also shows elevated σ values, suggesting interactions with the other parameters or nonlinear behaviour.

The replication of the buffer pixel and temperature delta findings from (Tonolla and Antonetti, 2026) within the present work, using an independently implemented Morris experiment on a different dataset, lends additional credibility to both sets of results and suggests that these sensitivity patterns are a robust property of the CWP detection algorithm rather than artefacts of a particular study context.

6.5 Answering Research Question 2

Research question 2 asked what adjustments could be made to the detection algorithm to reduce variability with respect to its parameters while also achieving plausible detection results. Based on the Morris screening findings, three modifications were implemented in v3, as described in Chapter 5: the buffer pixel count was fixed by factor fixing, the slab-based reference temperature computation was replaced by an IDW scheme removing the step length dependency, and the temperature delta parameter was retained as a user-defined input. The evaluation presented in Section 5.5 shows that v3 produces outputs comparable to v1 in terms of Jaccard similarity, while qualitatively eliminating the slab-boundary cutoff artefacts identified as a direct consequence of step-length sensitivity. Research question 2 can therefore be answered affirmatively: the identified parameter sensitivities can be addressed through targeted algorithmic modifications without sacrificing output plausibility.

6.6 Quellenverzeichnis

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